Name:VIGNA SUNDARI I
Reg No:23IT169

ASSIGNMENT – 1.2

Social Media Data Collection, Analysis, and Visualization

Introduction:

Social media platforms generate large volumes of data through user interactions such as posts, comments, likes, and shares. Analyzing this data helps understand user behavior, engagement patterns, and content popularity. However, ethical data collection and proper handling of publicly available information are essential. This report presents a study on collecting, preprocessing, analyzing, and visualizing publicly available data from a social media platform using modern data collection tools.

Selected Social Network:

YouTube is a video-sharing social media platform where users upload videos and interact through views, likes, comments, and subscriptions. These engagement metrics are publicly visible for most channels, making YouTube suitable for data analysis.

Type of Data Collected:

From a public YouTube channel/profile, the following data can be collected:

- Video title
- View count
- Like count
- Comment count
- Upload date
- Video URL

All these metrics are publicly available.

Data Collection Method

Tool Used: Apify Instagram Scraper (Web Scraping Tool)

The data was collected using the Instagram Scraper tool available on the Apify platform. This tool extracts structured video-level data without complex API setup.

Justification for Choosing Apify Store

- No API key required
- Easy export as CSV
- Public data only
- Simple implementation
- Suitable for academic use

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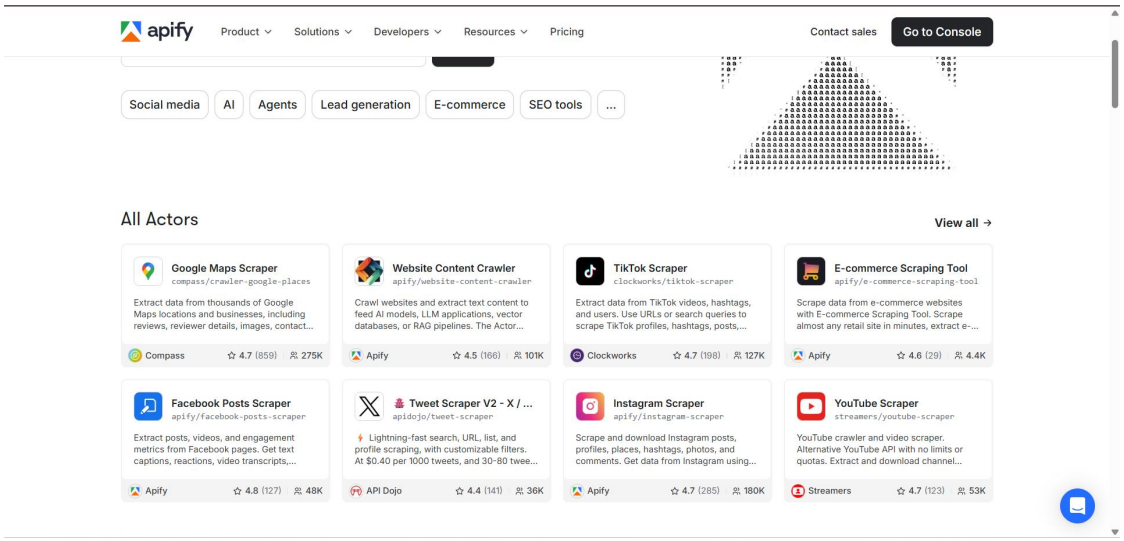
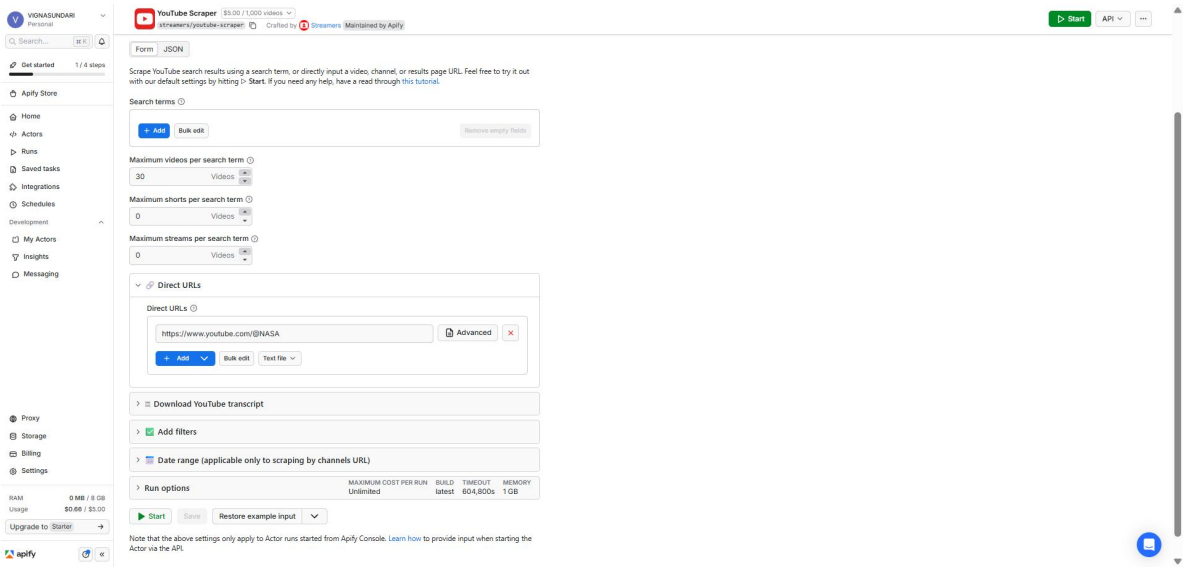


Implementation of Data Collection:

The following steps were performed:

1. Open Apify platform
2. Select “YouTube Scraper” actor
3. Enter YouTube channel URL
4. Configure to scrape videos
5. Select fields:
    - Title
    - viewCount
    - likeCount
    - commentCount
    - uploadDate

Export as CSV



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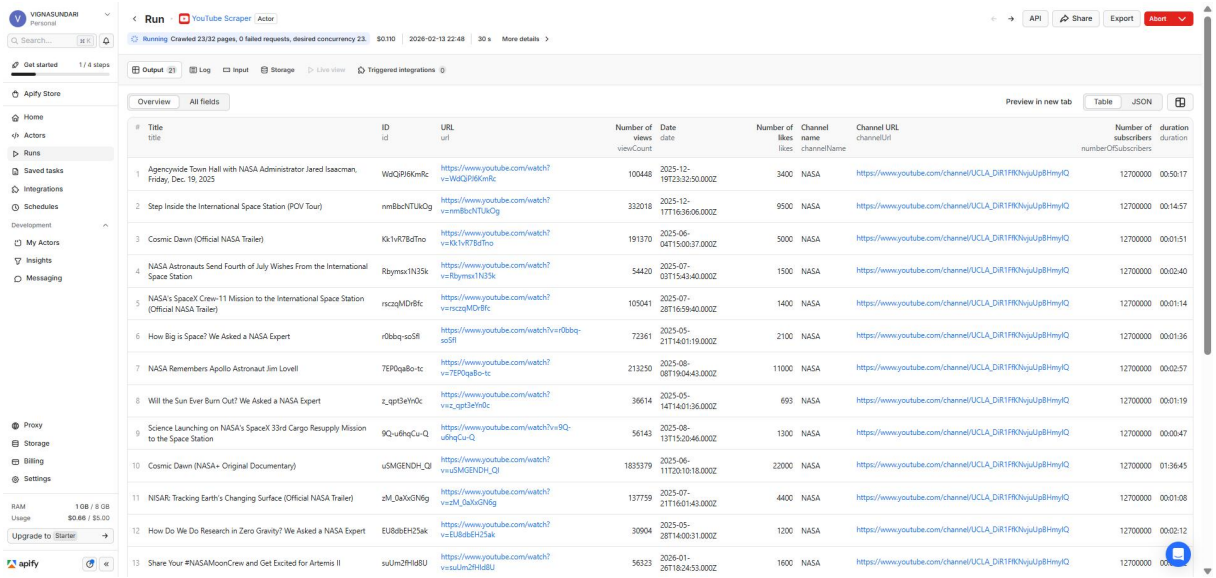
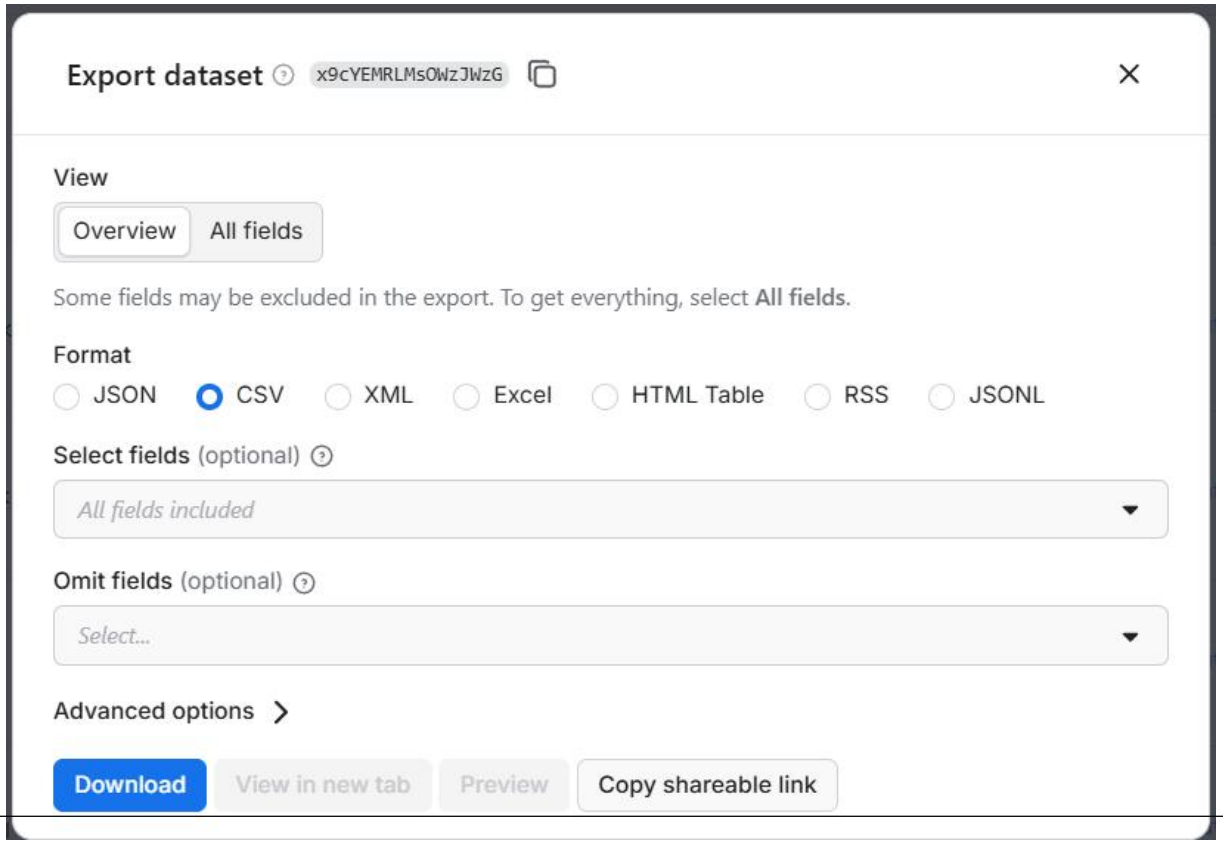


#	Title title	ID id	URL url	Number of views viewCount	Date date	Number of likes likes	Channel name channelName	Channel URL channelUrl	Number of subscribers numberOfSubscribers	duration duration
1	Agencywide Town Hall with NASA Administrator Jared Isaacman, Friday, Dec. 19, 2025	WdQiPj6KmRc	https://www.youtube.com/watch?v=WdQiPj6KmRc	100448	2025-12-19T23:32:50.000Z	3400	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:50:17
2	Step Inside the International Space Station (POV Tour)	nmBbcNTUkOg	https://www.youtube.com/watch?v=nmBbcNTUkOg	332018	2025-12-17T16:36:06.000Z	9500	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:14:57
3	Cosmic Dawn (Official NASA Trailer)	Kk1vR7BdTno	https://www.youtube.com/watch?v=Kk1vR7BdTno	191370	2025-06-04T15:00:37.000Z	5000	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:01:51
4	NASA Astronauts Send Fourth of July Wishes From the International Space Station	Rbymsx1N35k	https://www.youtube.com/watch?v=Rbymsx1N35k	54420	2025-07-03T15:43:40.000Z	1500	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:02:40
5	NASA's SpaceX Crew-11 Mission to the International Space Station (Official NASA Trailer)	rsczqMDrBfc	https://www.youtube.com/watch?v=rsczqMDrBfc	105041	2025-07-28T16:59:40.000Z	1400	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:01:14
6	How Big is Space? We Asked a NASA Expert	r0bbq-so5fI	https://www.youtube.com/watch?v=r0bbq-so5fI	72361	2025-05-21T14:01:19.000Z	2100	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:01:36
7	NASA Remembers Apollo Astronaut Jim Lovell	7EP0qaBo-tc	https://www.youtube.com/watch?v=7EP0qaBo-tc	213250	2025-08-08T19:04:43.000Z	11000	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:02:57
8	Will the Sun Ever Burn Out? We Asked a NASA Expert	z_qpt3eYn0c	https://www.youtube.com/watch?v=z_qpt3eYn0c	36614	2025-05-14T14:01:36.000Z	693	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:01:19
9	Science Launching on NASA's SpaceX 33rd Cargo Resupply Mission to the Space Station	9Q-u6hqCu-Q	https://www.youtube.com/watch?v=9Q-u6hqCu-Q	56143	2025-08-13T15:20:46.000Z	1300	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:00:47
10	Cosmic Dawn (NASA+ Original Documentary)	uSMGENDH_QI	https://www.youtube.com/watch?v=uSMGENDH_QI	1835379	2025-06-11T20:10:18.000Z	22000	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	01:36:45
11	NISAR: Tracking Earth's Changing Surface (Official NASA Trailer)	zM_0aXxGN6g	https://www.youtube.com/watch?v=zM_0aXxGN6g	137759	2025-07-21T16:01:43.000Z	4400	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:01:08
12	How Do We Do Research in Zero Gravity? We Asked a NASA Expert	EU8dbEH25ak	https://www.youtube.com/watch?v=EU8dbEH25ak	30904	2025-05-28T14:00:31.000Z	1200	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00:02:12
13	Share Your #NASAMoonCrew and Get Excited for Artemis II	suUm2fHld8U	https://www.youtube.com/watch?v=suUm2fHld8U	56323	2026-01-26T18:24:53.000Z	1600	NASA	https://www.youtube.com/channel/UCLA_DiR1FfKNvjuUpBHmylQ	12700000	00



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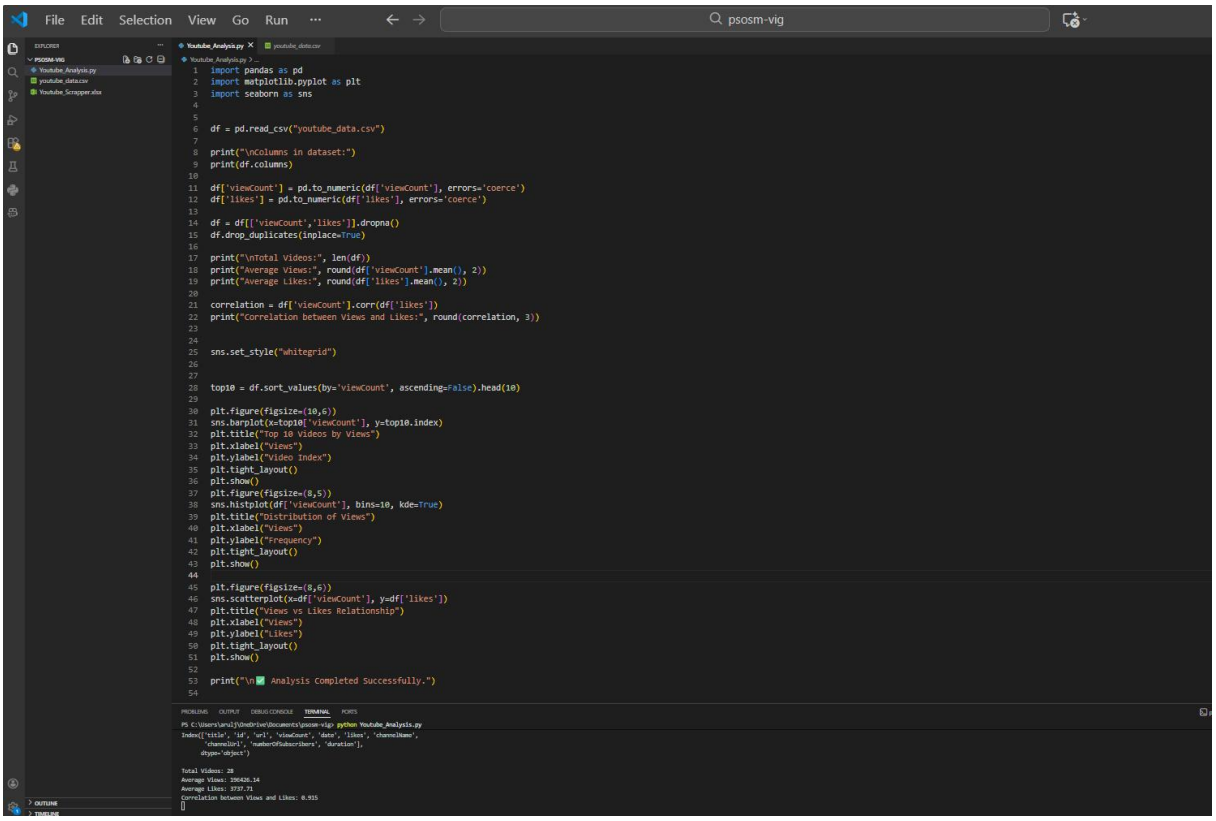
Data Storage and Preprocessing

After downloading the CSV file:

Preprocessing Steps:

1. Load dataset using Pandas
2. Convert viewCount, likeCount, commentCount to numeric
3. Remove missing values
4. Remove duplicates
5. Select relevant engagement columns



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```
PS C:\Users\arulj\OneDrive\Documents\psosm-vig> python Youtube_Analysis.py
Columns in dataset:
Index(['title', 'id', 'url', 'viewCount', 'date', 'likes', 'channelName',
       'channelUrl', 'numberOfSubscribers', 'duration'],
      dtype='object')

Total Videos: 28
Average Views: 196426.14
Average Likes: 3737.71
Correlation between Views and Likes: 0.915
```

Data Analysis and Visualization

The cleaned YouTube dataset was analyzed using Python libraries including:

- Pandas
- Matplotlib
- Seaborn

Analysis Performed

- Total number of videos collected
- Average views per video
- Average likes per video
- Average comments per video
- Identification of most engaging videos
- Relationship between views and likes
- Distribution of engagement levels

Visualizations Used

- Bar charts were used to identify the top-performing videos and compare view counts across content. This helped highlight which videos attracted the highest audience attention.
- A histogram was used to analyze the overall distribution of video views, revealing whether engagement was evenly distributed or concentrated among a few popular videos.
- A scatter plot was used to examine the relationship between views and likes, showing how audience interaction increases with video popularity.

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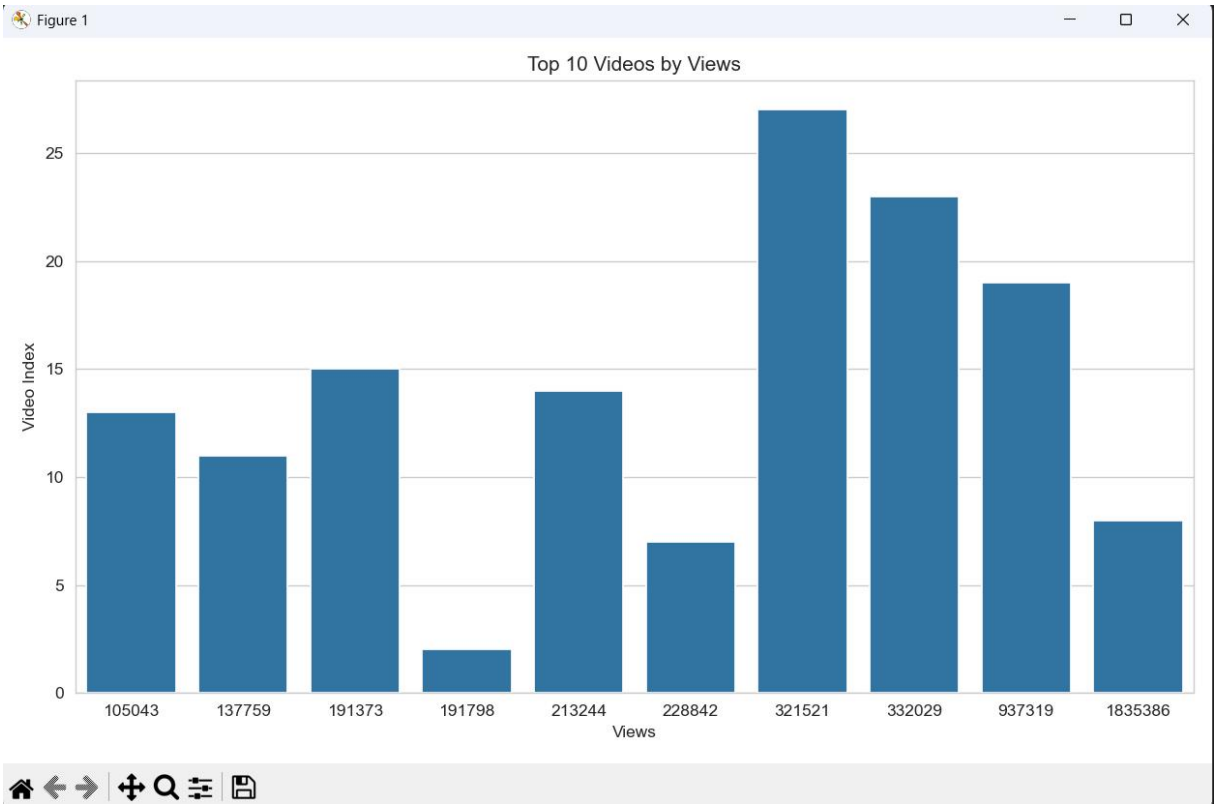

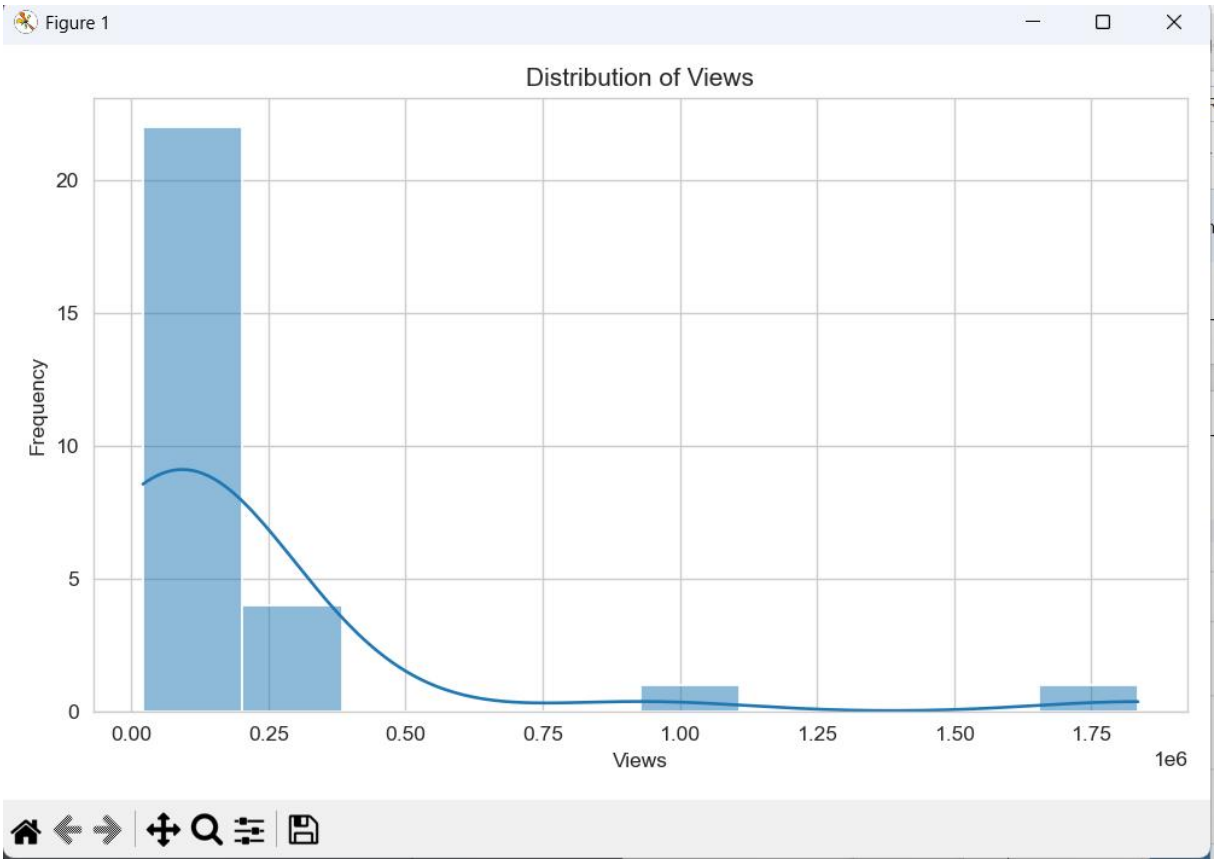

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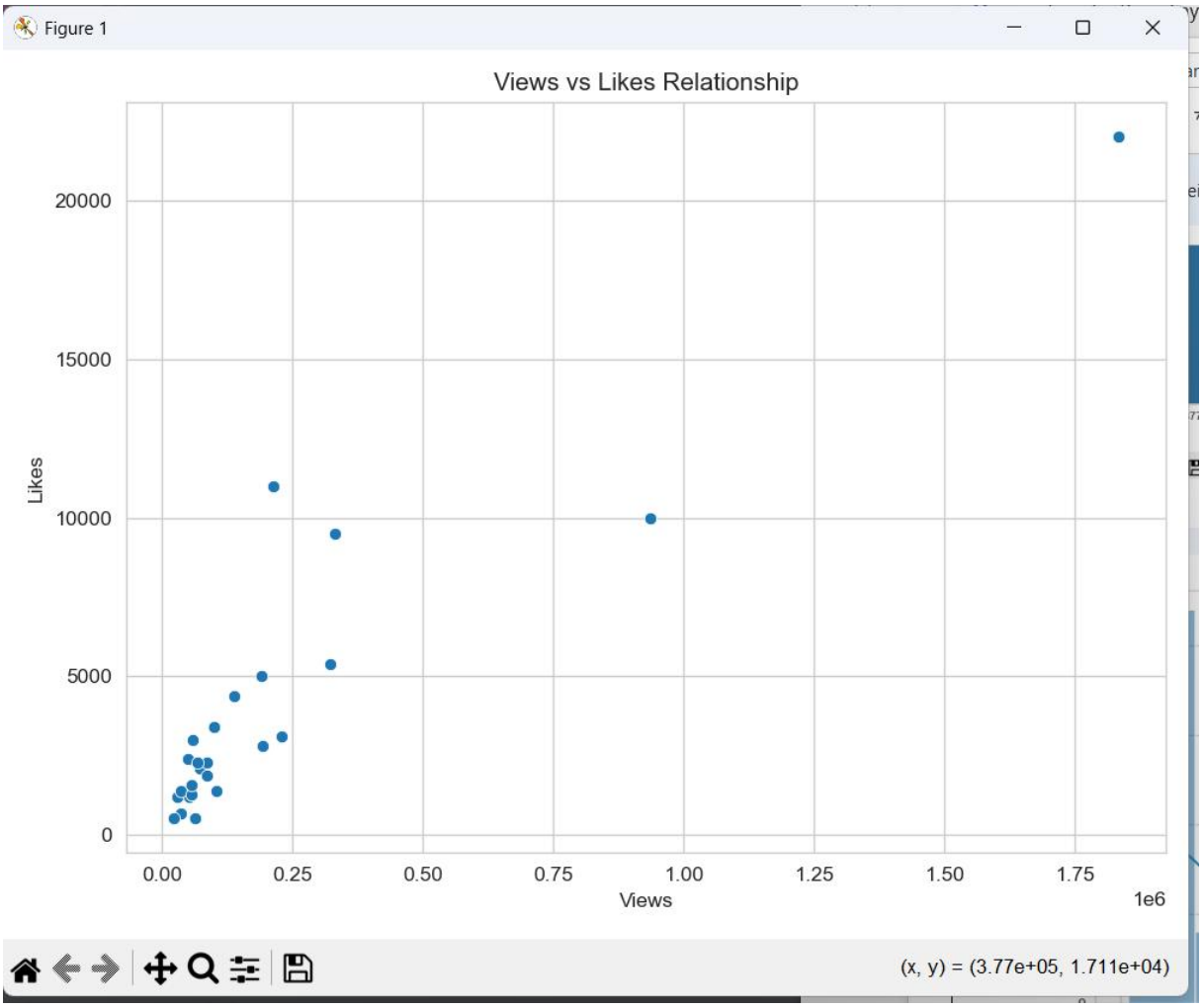

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Observations

- The bar chart showed that a few videos received significantly higher views, indicating that engagement is concentrated among top-performing content.
- The histogram revealed a positively skewed distribution, where most videos had moderate views and only a few achieved very high popularity.
- The scatter plot demonstrated a positive relationship between views and likes, showing that higher-viewed videos generally receive more audience engagement**.**

Ethical Considerations

- This study strictly followed ethical guidelines during data collection and analysis.
- Only publicly available information from the selected YouTube channel was collected.
- No private accounts, restricted content, or personal user data were accessed.
- The data was used solely for academic purposes and not for commercial or redistribution activities.
- All data collection was conducted using authorized scraping tools while respecting platform policies and responsible data handling practices.

Conclusion

This study successfully demonstrated the process of collecting, preprocessing, analyzing, and visualizing publicly available YouTube engagement data. The analysis revealed that video engagement is unevenly distributed, with a small number of videos receiving significantly higher views and interactions. A positive relationship was observed between views and likes, indicating that popular videos tend to generate higher audience engagement. The visualizations provided meaningful insights into content performance and user interaction patterns.

Future Scope

Future improvements to this study may include:

- Performing sentiment analysis on video comments to understand audience opinions
- Conducting time-based analysis to identify engagement trends over months or years
- Comparing multiple YouTube channels for competitive analysis
- Applying machine learning techniques to predict video engagement
- Analyzing hashtag usage and video categories for deeper content insights